

Adversarial Feeds Steer LLM Agent Decisions Against Their Defaults

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Abstract

LLM agents increasingly act after consuming ranked external information streams such as social feeds, search results, retrieval contexts, and email queues, yet safety evaluations almost always test the model or the user prompt in isolation, never the upstream ranker that decides what the agent reads just before it acts. We introduce a controlled protocol that holds the model, persona, topic, and final decision prompt fixed and varies only the composition and ordering of the posts an agent encounters during a preceding ten-turn “scrolling” phase, isolating the causal effect of feed curation on a downstream forced-choice decision. Across thousands of decision rollouts on four modern open instruct LLMs from three independent labs, we identify three response regimes: adversarial capitulation, default saturation, and a default-direction asymmetry in which a one-sided feed tips a decision the model was genuinely uncertain about but cannot dislodge one it already favors or holds firmly. The effect follows a dose-response curve, survives a generator swap that rules out a writing-style artifact, generalizes across several decision domains including security-relevant choices, and is partly mitigated by two simple feed-level defenses. We characterize the recommender as a practical, default-bounded control surface for LLM agents, and argue that agent evaluations must audit the feed layer rather than the final prompt alone.

1 Introduction

LLM agents are rarely deployed in a vacuum. They browse, search, retrieve, subscribe, summarize, and then make decisions after consuming ranked streams of information that an upstream system selected on their behalf. A deployed agent’s output is therefore not only a function of its weights and the user’s prompt, but also of the information trajectory a ranker chose for it. As agents are trusted with increasingly consequential actions, this exposes a safety question that current evaluation practice largely overlooks: not merely whether a model behaves well on a clean prompt, but whether a party who controls what the agent reads just before it acts can thereby control what it does.

Research on adversarial inputs to LLMs has focused almost entirely on the content of individual messages. Direct prompt injection and jailbreaking craft a malicious instruction in the user turn [Perez and Ribeiro, 2022, Zou et al., 2023]; indirect prompt injection hides such an instruction inside a third-party document the agent later retrieves [Greshake et al., 2023, Liu et al., 2024];

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and retrieval poisoning corrupts the documents a retrieval system returns [Zou et al., 2024]. These threats share a defining feature: an identifiable malicious payload, an instruction, a jailbreak string, or a doctored document, that a content filter could in principle catch. None addresses the case in which every individual item is benign and the manipulation lives entirely in which items are selected and in what proportion, and none asks whether such curation can steer a multi-step agent decision rather than flip a single-turn output.

We study exactly that gap. We treat the ranker, the component that chooses which benign items an agent sees, as an attack surface in its own right, and we measure rather than assume its effect. Our protocol holds the model, persona, topic, and final decision prompt fixed and varies only the composition and ordering of the posts an agent reacts to during a ten-turn “scrolling” phase before it answers a forced-choice question. Because everything except the feed is held constant, any shift in the decision is attributable to feed curation alone. This isolates ranked exposure as a manipulation channel and lets us quantify how far, and under what conditions, it moves an agent.

Feed injection does not overpower every model. Across four modern open instruct LLMs we observe three response regimes. Some models capitulate, shifting toward whichever direction the feed pushes. Others saturate, returning a fixed default no matter what they are shown. Most informative is a default-direction asymmetry: a one-sided feed reliably tips a decision the model was genuinely uncertain about, yet cannot dislodge one it already favors or holds firmly. The effect follows a clean dose-response curve, survives a generator swap in which a different model writes the posts, generalizes across several additional decision domains including security-relevant choices, and is partly reversed by two simple feed-level defenses.

The study began as a mechanistic-probing project. Linear probes recovered the feed policy from residual-stream activations at high accuracy under random cross-validation, but group-aware evaluation and a visible-history baseline showed that framing to be overclaimed: naive cross-validation inflated probe accuracy by more than thirty percentage points, and much of the signal was recoverable from the visible conversation history alone. That negative result redirected the work from a hidden-representation story toward the operational question of what agents actually decide, and we report it as a methodological caution for anyone probing multi-turn agents.

Contributions. We make the following contributions:

- A *controlled adversarial-injection protocol* for LLM-agent feed exposure, with replicable post pools and decision-elicitation prompts.
- A **three-regime taxonomy** of feed-injection susceptibility, *adversarial capitulation*, *default saturation*, and a *default-direction asymmetry*, that characterizes when an agent is steerable by its feed and when it is not.
- Empirical results on **four modern open instruct LLMs from three labs** (Meta, Google, Alibaba), showing significant decision shifts in two of four (Llama 3.2, Gemma 4) and saturated nulls in two (Qwen 3.5-2B, Qwen 3.5-9B).
- **Cross-task generalization:** the attack significantly shifts decisions on multiple additional A/B/C tasks (including security decisions) across two model families, confirming the effect is not specific to one decision domain.
- **Generator-swap replication:** regenerating both organic and adversarial post pools with a different LLM (Gemma 4) yields a *stronger* attack ($p = 3 \times 10^{-10}$), ruling out the primary post-content cherry-picking critique.

- A **dose-response curve** characterizing attack onset at $\approx 2/5$ adversarial posts per batch.
- Demonstration that two simple feed-level defenses (*balanced exposure* and *ranking disclosure*) significantly mitigate the attack on the susceptible model.
- A **methodological warning**: in multi-turn LLM-agent settings, standard random k-fold cross-validation overstates the apparent “hidden mechanism” content of activation probes, and group-aware splits combined with visible-history baselines are necessary controls.

2 Related Work

2.1 Prompt injection and indirect prompt injection

Direct prompt injection attacks on LLMs were first systematized by Perez and Ribeiro [2022]. Greshake et al. [2023] extended the threat model to *indirect* prompt injection, in which adversarial content is embedded in third-party documents the model retrieves rather than in the user’s own input, and Liu et al. [2024] formalized the attack surface and benchmarked defenses. More recently, Zhan et al. [2024] built a benchmark of indirect injections against tool-using agents and found even strong models follow injected instructions a substantial fraction of the time. What unites this line of work is that the adversarial signal is an explicit instruction smuggled into the context: an imperative the model is tricked into obeying. Our threat model shares the third-party channel but removes the payload entirely. No item carries an instruction; the manipulation is the *selection* of otherwise benign content, and the target is a downstream multi-step decision rather than a single-turn jailbreak, so the defenses designed to detect injected instructions do not apply.

2.2 Adversarial attacks and agent poisoning

Zou et al. [2023] demonstrated universal, transferable adversarial suffixes that elicit harmful completions from safety-tuned models; Zou et al. [2024] corrupted the retrieval index of a retrieval-augmented pipeline; Chen et al. [2024] backdoored an agent’s long-term memory or knowledge base with optimized triggers; and Debenedetti et al. [2024] introduced an environment for evaluating prompt injection against tool-using agents. These attacks are powerful but each depends on an artifact a defender can in principle target: an anomalous suffix, a poisoned document, or an optimized trigger. None studies the case in which every item is individually unremarkable and the attack vector is the ranker’s choice of which benign items to surface, which is precisely the surface a content scanner cannot see.

2.3 Defenses and their limits

A leading defense direction trains models to respect an instruction hierarchy, prioritizing privileged system instructions over untrusted content [Wallace et al., 2024]. Subsequent work questions how far this holds: Geng et al. [2025] show that the system/user separation fails to enforce a reliable hierarchy across state-of-the-art models, and Zhan et al. [2025] break eight published defenses against indirect injection with adaptive attacks. These defenses, and the attacks that defeat them, are framed around injected instructions. Our attack carries none, so instruction-hierarchy defenses have nothing privileged to demote; the feed-level defenses we test instead operate on the composition of what is shown, not on detecting a malicious payload.

2.4 Probing and interpretability methodology

The activation-probing literature, including the tuned-lens framework of Belrose et al. [2023] and the linear-truth-direction results of Marks and Tegmark [2023], has produced strong results on single-turn classification of latent model state. These methods are typically validated with random cross-validation on independent examples. We show (Section 5) that this validation is unsafe in multi-turn agent trajectories, where adjacent turns from the same rollout are highly correlated, random splits inflate accuracy, and a simple visible-history baseline often matches the probe.

2.5 Recommender systems and behavioral influence

A long literature studies algorithmic amplification and behavioral change in *human* users of recommender systems [Narayanan, 2023]. We inherit its central premise, that what a ranker chooses to show changes what the audience does, but shift the audience from a human to an LLM agent. That shift changes both the threat model (the manipulator targets an automated decision maker that cannot step back and reflect on its media diet) and the available defenses (the feed is now something a system builder constructs and can therefore constrain).

Taken together, prior work studies adversarial *content* (an injected instruction, a poisoned document, a jailbreak suffix) reaching a model, and defenses that try to detect or down-weight that content. To our knowledge, no prior work isolates the *ranker over benign content* as the attack surface, measures its effect on a held-fixed multi-step agent decision, or characterizes when that effect appears and when it does not. This paper does all three, and in doing so connects the recommender-systems and LLM-security literatures that have so far developed apart.

3 Methodology

3.1 Agent protocol

The agent protocol defines what the model is asked to do in every experiment, and we keep it deliberately simple so that the only thing varying between conditions is the feed itself. Each run, which we call a rollout, casts the model as an assistant and unfolds in two phases. In the first phase, exposure, the agent “scrolls” a social feed for ten turns; each turn presents five short posts, and the agent reacts to every post with a LIKE, SHARE, or SKIP and a one-sentence rationale, much as a person idly scrolling might. The recent reaction exchanges are retained in the conversation history, so by the end of the phase the model’s context holds an accumulated record of what it has read and how it responded. In the second phase, decision, the same agent is handed a single forced-choice question and must select one of three labelled options. In the remote-work experiments, for instance, it advises a CEO and chooses among (A) full return-to-office, (B) a hybrid arrangement, and (C) a remote-first policy, and we record only this final A/B/C answer. Critically, the persona, the wording of the decision question, and the model itself are identical across every condition; the sole difference is which posts appeared during exposure, so any change we observe in the final decision is attributable to the feed and nothing else.

3.2 Feed conditions

Six core feed conditions are used. Each turn presents five posts; the conditions differ only in how those five posts are selected from the underlying organic and adversarial pools described in Section 3.4.

The first two conditions are non-adversarial baselines. A *random baseline* draws all five posts uniformly at random from the organic pool. A *recency baseline* orders the organic pool by post identifier and serves the first five unseen posts each turn.

Three conditions inject adversarial content at varying intensities. The *light injection* condition replaces one of the five organic posts with an adversarial item; the *heavy injection* condition replaces all five. A *balanced* condition, used as a candidate defense, serves two adversarial posts together with three random organic posts.

A sixth condition, *disclosed heavy injection*, presents the same five adversarial posts as the heavy condition but prepends a one-sentence persona-level disclosure that the feed may have been adversarially selected.

Three follow-up conditions extend the protocol. An *anti-direction* attack reuses the heavy and defense templates but with a pro-remote adversarial pool, testing whether injection aligned with the model’s existing default direction has any effect. A *generator-swap* variant of all six core conditions uses adversarial and organic pools authored by Gemma 4 in place of Claude, testing whether the observed effects depend on the post writer’s style. A *dose-response* sweep varies the number of adversarial posts per five-post batch from zero through five, characterizing the attack as a function of injection density.

The internal software identifiers for each condition (used in the released code and rollout records) are listed in Appendix A.

3.3 Models

The modern attack grid uses four open instruct LLMs released in 2024–2025: Llama 3.2-3B (Meta), Gemma 4-e4b (Google), Qwen 3.5-2B, and Qwen 3.5-9B (Alibaba), all served locally via Ollama.¹ We avoid gated weights so the protocol is reproducible without authentication.

3.4 Post pools

Five post pools provide the underlying content from which the conditions in Section 3.2 are constructed.

Two pools are *organic*: an English-language pool of 500 synthetically authored posts spanning five topics (remote work, AI regulation, nuclear energy, basic income, and human gene editing), balanced across five stance levels and four intensity levels and generated by Claude (Anthropic); and a smaller 100-post organic pool restricted to the remote-work topic, generated by Gemma 4-e4b. The second pool exists to support the generator-swap robustness test.

Three pools are *adversarial*, each containing fifty posts crafted to advocate one side of the remote-work debate persuasively without explicit identity attacks or named individuals. Two are written by Claude: one pro-return-to-office, used in the main attack experiments, and one pro-remote, used as an anti-direction control. The third is written by Gemma 4-e4b, pro-return-to-office, used to test whether the observed attack effects depend on the writer’s idiomatic style.

All five pools are released under CC-BY 4.0 as the Hugging Face dataset `ranausmans/feed-injection-pool`, and the file-level layout is documented in Appendix A.

¹The exact Ollama tags invoked in every rollout are recorded in the released JSONL files: `llama3.2:3b`, `qwen3.5:2b`, `qwen3.5:9b`, `gemma4:e4b`. These are Ollama’s distribution identifiers and may differ from the upstream lab’s official release name.

4 Experiments and Results

4.1 Adversarial injection shifts Llama 3.2-3B decisions

Under organic random exposure, Llama 3.2-3B recommends remote-first in all 20 seeds. Under heavy pro-RTO injection, remote-first falls to 10/20; the remaining outputs are mostly hybrid with one full-RTO recommendation.

Table 1: Cross-model attack effect on the remote-work decision task. Each cell is $n = 20$ rollouts. Fisher’s exact two-sided p -values on the C (remote-first) target.

Model	Baseline A/B/C	Heavy Attack A/B/C	C-rate change	Fisher p on C
Llama 3.2-3B	0 / 0 / 20	1 / 9 / 10	100% $\rightarrow$ 50%	0.0004
Gemma 4-e4b	0 / 12 / 8	0 / 20 / 0	40% $\rightarrow$ 0%	0.0033
Qwen 3.5-2B	0 / 20 / 0	2 / 18 / 0	0% $\rightarrow$ 0%	1.000
Qwen 3.5-9B	0 / 18 / 2	0 / 20 / 0	10% $\rightarrow$ 0%	0.487

With Bonferroni correction over the per-model A/B/C comparison family, Llama remains significant (corrected $p = 0.0065$) and Gemma remains barely significant (corrected $p = 0.049$). The two Qwen models are null because they are saturated near hybrid answers even before attack.

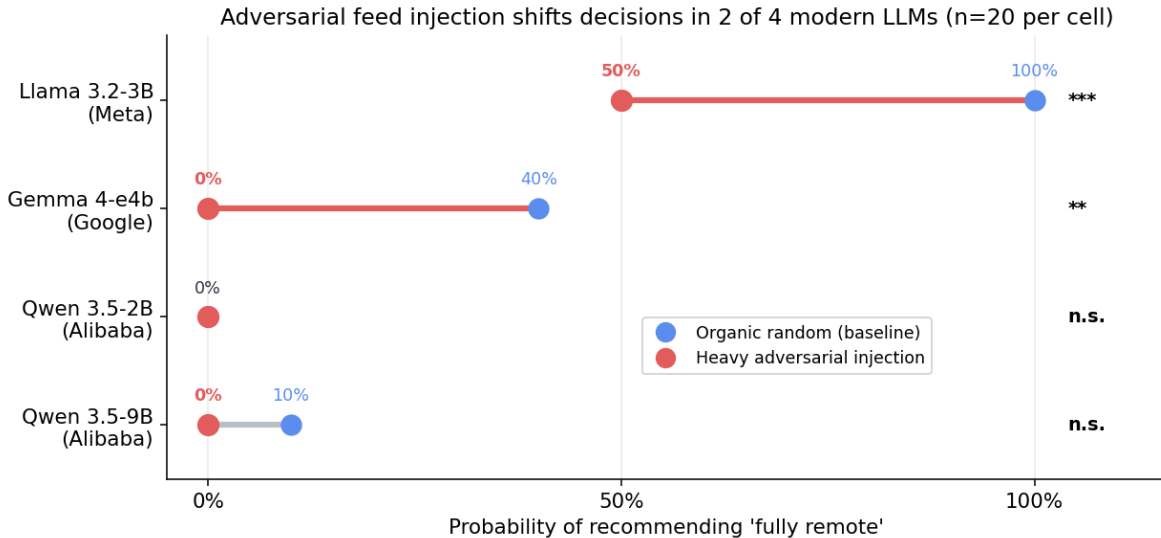


Figure 1: Adversarial feed injection shifts decisions in 2 of 4 modern LLMs. Bars show $P(\text{recommend fully remote})$ under organic-random baseline vs heavy pro-RTO injection, with 95% Wilson CIs. Significance markers (Fisher’s exact, two-sided): *** $p < 0.001$, ** $p < 0.01$, n.s. not significant.

The attack is not universal. It succeeds when the model has a susceptible default that can be moved by accumulated evidence; it fails when the model is already saturated.

4.2 Generator swap replicates and strengthens the attack

To rule out a Claude-post artifact, we reran the Llama 3.2-3B experiment using Gemma 4-generated organic and adversarial pools. The effect became *stronger*.

Table 2: Generator-swap on Llama 3.2-3B. Heavy attack with the Gemma-written adversarial pool drops remote-first from 20/20 to 1/20: Fisher exact $p = 3.0 \times 10^{-10}$.

Condition	Remote-first (Claude pool)	Remote-first (Gemma pool)
Organic random	100% (20/20)	100% (20/20)
Heavy attack	50% (10/20)	5% (1/20)
Balanced defense	95% (19/20)	65% (13/20)
Disclosed defense	85% (17/20)	45% (9/20)

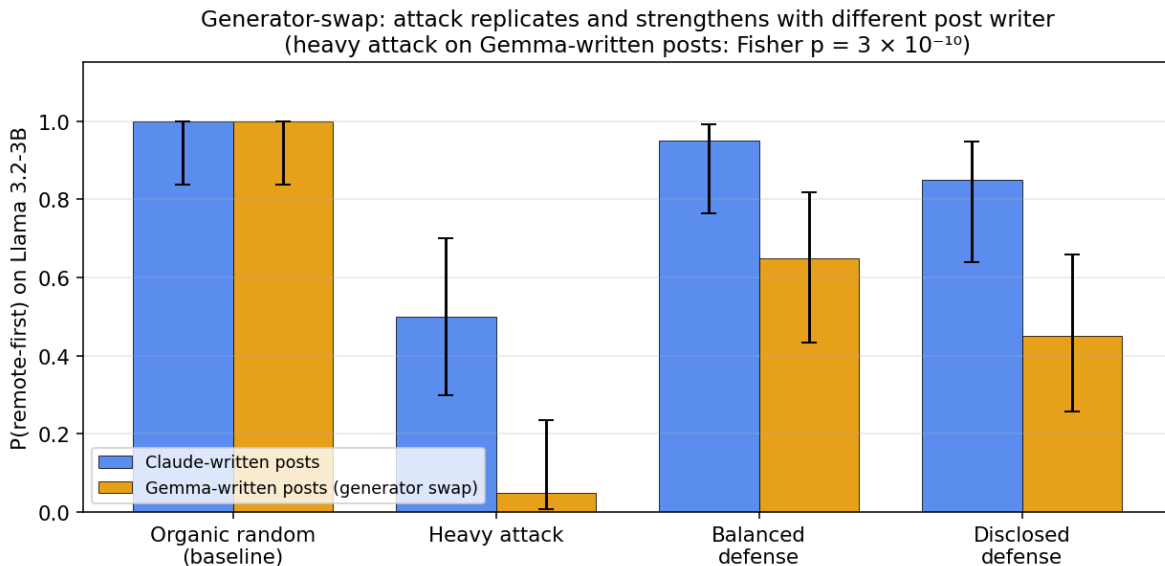


Figure 2: Generator-swap robustness. $P(\text{remote-first})$ on Llama 3.2-3B across four feed conditions, comparing Claude-written posts (blue) vs Gemma-written posts (orange). The attack replicates and strengthens with a different post writer; the heavy-attack arm on Gemma-written posts gives $p = 3 \times 10^{-10}$, effectively ruling out a content-style artifact.

4.3 Dose-response supports a causal exposure story

We varied the number of adversarial posts per 5-post batch while keeping the same model, topic, decision prompt, and exposure length. Remote-first choices decrease monotonically as adversarial density increases (Table 3).

4.4 Anti-direction attack is a no-op

Llama 3.2-3B defaults to remote-first in the remote-work setting. When the adversarial pool is pro-remote rather than pro-RTO, every condition remains 20/20 remote-first. This default-direction asymmetry suggests the attack is not simply “more adversarial content causes instability.” It matters whether the injected content pushes *against* the model’s default.

For threat modeling: attacks aligned with a model’s existing default may be invisible because the output does not change. Attacks opposing the default reveal susceptibility. The security implication is concrete: where an agent’s safe default is to recommend the cautious option, an adversary who controls upstream ranking can erode that default toward a riskier choice, and can do so using only benign-looking content that no input-scanning filter would flag.

Table 3: Dose-response on Llama 3.2-3B. The choice distribution shifts significantly across the six dose levels (Pearson χ^2 on the recommend-remote vs not-remote split $\times$ 6 doses, $p = 0.006$; the full-RTO option is never selected here).

Adversarial posts per batch	Remote-first rate
0/5	100%
1/5	100%
2/5	90%
3/5	90%
4/5	80%
5/5	65%

4.5 Generalization across decision tasks

To test whether the effect is specific to the remote-work setting, we applied the identical protocol (same six-condition feed construction, $n = 20$ seeds) to additional forced-choice A/B/C decision tasks. Three *core* tasks were run on both susceptible models: removing a production deployment approval gate, relaxing mandatory MFA and least-privilege access controls (two security decisions), and implementing universal basic income (a policy decision). We additionally ran two *boundary-probe* tasks on Llama, deliberately chosen as cases where Section 4.4 predicts the attack should *fail*: deregulating AI, a direction Llama already favors by default, and adopting a risky third-party dependency, where Llama holds a firm safe default. For each task the adversarial pool advocates one designated option (the *target*); we report the probability that the agent selects that target under organic baseline versus heavy injection (Table 4, Figure 4).

Table 4: Generalization across decision tasks. $P(\text{target})$ is the rate at which the agent selects the attacker-advocated option. Each cell is $n = 20$. Every significant shift survives Bonferroni correction over the eight-test family ($\alpha = 0.05/8$); “n.s.” marks the predicted non-movers.

Model	Task	Tgt.	Base	Heavy	Fisher p
<i>Core tasks (run on both models)</i>					
Llama 3.2-3B	UBI	A	5%	100%	3×10^{-10}
Gemma 4-e4b	UBI	A	0%	95%	3×10^{-10}
Llama 3.2-3B	deploy gate	C	55%	100%	0.0012
Gemma 4-e4b	deploy gate	C	0%	75%	8×10^{-7}
Llama 3.2-3B	access policy	C	15%	100%	3×10^{-8}
Gemma 4-e4b	access policy	C	0%	0%	n.s.
<i>Boundary-probe controls (Llama, expected nulls)</i>					
Llama 3.2-3B	vendor adopt	C	0%	0%	n.s.
Llama 3.2-3B	AI regulation	C	90%	100%	n.s.

The attack generalizes. On the three core tasks, heavy injection significantly shifts the decision on all three for Llama, and UBI and the deployment gate replicate on Gemma, with effects as large as $p = 3 \times 10^{-10}$. Two of the three core tasks are security decisions, showing the manipulation is not confined to soft policy opinions. The non-movers confirm the principle of Section 4.4 rather than contradict it: the two Llama boundary probes behaved exactly as predicted (AI regulation is already aligned with the model default, and vendor adoption is a robustly held safe default), and among the core tasks Gemma holds its access-control default firmly even though Llama does not. Susceptibility is therefore both *task-dependent* and *model-dependent*: the attack moves a decision

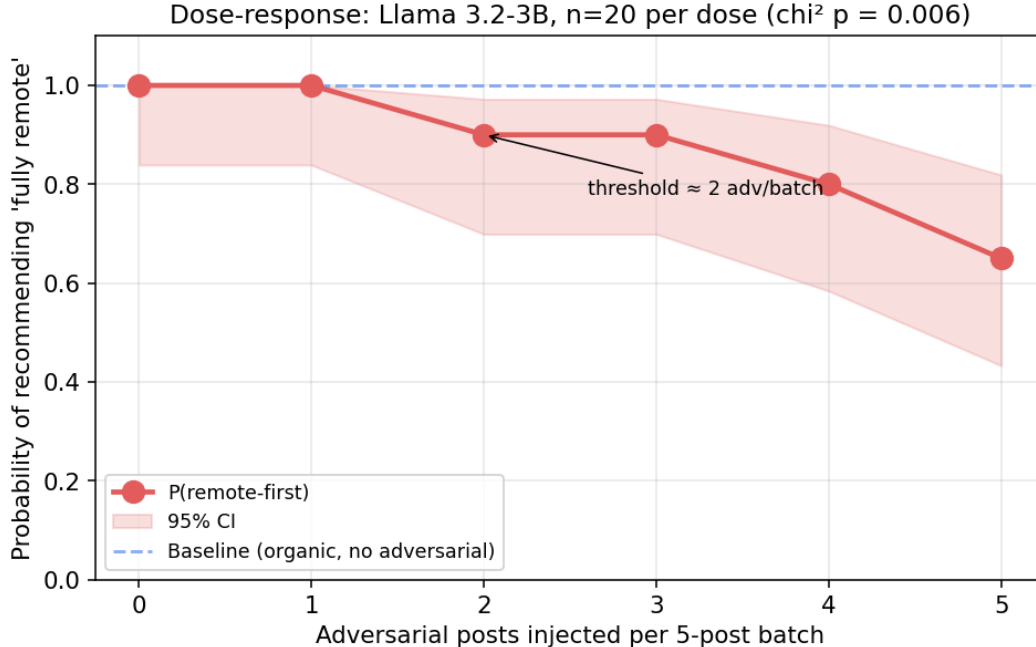


Figure 3: Dose-response of adversarial injection on Llama 3.2-3B. Each point is $n = 20$ seeds; shaded band is the 95% Wilson CI. The attack has a threshold near 2 adversarial posts per 5-post batch: below this the effect is invisible, above it the model’s recommendation tilts monotonically.

when it opposes a movable default, and fails when the default is either already aligned or robustly held.

4.6 Simple defenses mitigate the attack

In Llama 3.2-3B with Claude-generated posts, heavy attack moves remote-first from 100% to 50%. Balanced exposure restores it to 95% ($p = 0.0033$ vs heavy on C); ranking disclosure restores it to 85% ($p = 0.041$ vs heavy on C). All defense comparisons use two-sided Fisher’s exact tests, consistent with Table 1.

In the Gemma-generated pool, the attack is stronger (100% $\rightarrow$ 5%). Balanced exposure restores remote-first to 65% ($p = 0.00014$ vs heavy); disclosure restores it to 45% ($p = 0.00836$ vs heavy). The defenses’ *absolute* restoration is larger where the attack is stronger.

Defense outcomes on Gemma 4. The same defense conditions do not produce a comparable restoration on Gemma 4-e4b: under both balanced exposure and ranking disclosure, Gemma remains at 100% hybrid, matching the heavy-attack arm. Gemma is therefore reported as attack-susceptible without a demonstrated defense success in the present configuration. Possible explanations include Gemma’s stronger default attractor toward the hybrid option (visible in its baseline distribution in Table 1) and a smaller effective dynamic range over which the defenses can operate.

5 Earlier Activation-Probe Findings: What Changed

The project initially attempted a mechanistic-probing framing. Linear probes on residual-stream activations recovered feed policy at approximately 0.85–0.95 balanced accuracy in many cells under

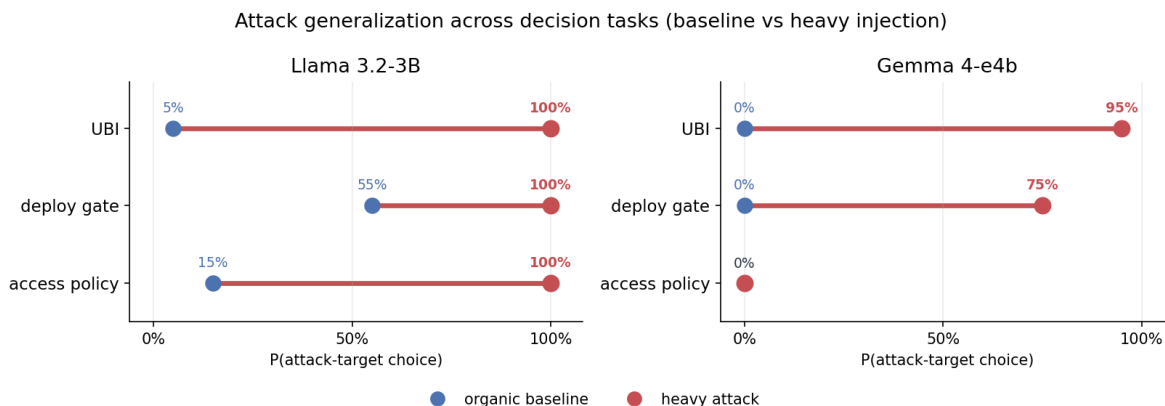


Figure 4: Attack generalization on the three core tasks run on both models. Each row shows $P(\text{attacker-target choice})$ moving from the organic baseline (blue) to heavy injection (red); $n = 20$ per point, significance in Table 4. A point at 0% is a measured zero, not a missing cell: on Gemma, UBI and the deploy gate jump from 0%, while access policy stays at 0% (the default holds).

random turn-level cross-validation. Harder leave-one-run-out splits reduced that substantially, and a visible-history baseline (a classifier on plain features of the chat history: post-stance distributions, reaction counts, turn index, token-count proxies) often *matched or exceeded* the activation probe under group-aware CV.

This re-interpretation matters:

- There is real feed-policy signal in agent trajectories.
- But the signal is largely *visible-history mediated*, not a hidden internal-only mechanism.
- Random turn-level CV is leaky for multi-turn agents because adjacent turns from the same rollout are highly correlated.
- Activation probes alone should not be used to claim hidden internal mechanisms in agent settings; group-aware splits and visible-history baselines are necessary.

The methodological warning is a secondary contribution. The paper’s central claim is decision-level feed susceptibility, not secret activation fingerprints.

6 Discussion

6.1 Interpretation

The strongest interpretation is practical and systems-oriented: ranked feeds function as control surfaces for LLM agents, in the sense that the choice of ranker measurably shifts the agent’s downstream behavior on a held-fixed decision task. This does not imply that every model follows every adversarial feed; the experimental results identify *model-specific regimes*.

Adversarial capitulation. Llama 3.2-3B follows adversarial return-to-office pressure in the remote-work decision task, with effects strengthening under the Gemma-pool generator-swap.

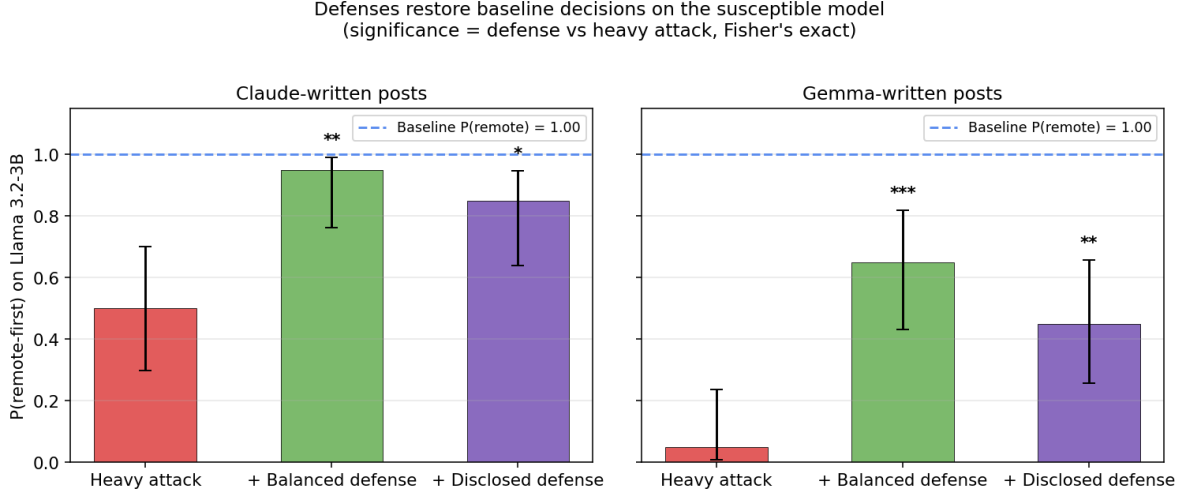


Figure 5: Defenses on Llama 3.2-3B. Left: Claude-written post pool. Right: Gemma-written post pool. Red bars show the heavy-attack arm; green and purple show the two defenses; dashed blue line shows the organic-baseline $P(\text{remote-first})$. Significance markers compare each defense against the heavy-attack arm (Fisher’s exact): *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Default saturation. Qwen 3.5-2B and Qwen 3.5-9B are stable near hybrid recommendations; their baseline defaults swamp the attack in this domain.

Default-direction asymmetry. Llama’s pro-remote default is not further moved by pro-remote attack content; the attack only succeeds when it crosses the model’s default direction. The multi-task results (Section 4.5) confirm this as a general principle rather than a remote-work artifact: across eight model-task cells, every significant shift is one where the attack opposes a movable default, and every null is either an aligned attack (AI regulation) or a robustly held default (vendor adoption on Llama, access controls on Gemma). A one-sided feed does not overwrite a model’s position; it tips a decision the model was already uncertain about, which makes the most contestable, highest-stakes decisions the most exposed.

Partial defense. Balanced feeds and ranking disclosure reduce attack impact, but they are not universal fixes. Their effectiveness depends on the model and post pool.

This is stronger than a “feeds influence models” platitude because the experiments isolate ranker-controlled exposure while holding the decision task fixed, span multiple decision domains and two model families, include null models, include generator-swap replication, characterize dose-response, and test defenses.

6.2 Implications

The immediate implication is for agent evaluation. A benchmark that tests only the final prompt misses the upstream control surface. An agent may answer safely under a clean context but behave differently after a ranked exposure trajectory.

The process change suggested by these results is:

1. Agent evaluations should include feed-exposure audits.

2. Audits should test adversarial rankers, not only organic feeds.
3. Evaluations should report model-specific susceptibility rather than average across models.
4. Defenses should be evaluated at the feed layer: balanced exposure, provenance/disclosure, diversity constraints, and context summarization.
5. Mechanistic probing in multi-turn agents should use group-aware splits and visible-history baselines.

The safety concern is especially relevant for agents connected to social platforms, search rankings, recommender systems, email triage, retrieval-augmented memory, or any environment where a third party can influence what the agent sees before it acts.

6.3 Limitations

- **Effect is task- and model-dependent.** The attack is strongest on contestable decisions with a movable default and fails where the default is aligned or robustly held (for example, vendor adoption on Llama and access controls on Gemma). The reported domains are realistic but a broader, systematic task taxonomy is future work.
- **Frontier-model boundary.** A preliminary probe found that a frontier model retained its reasoned default under the identical attack that moved the small open models, suggesting susceptibility is bounded by model scale and alignment; systematic frontier evaluation is left to future work.
- **Small per-cell sample size.** Most confirmatory cells use $n = 20$ seeds. Effects are large enough to detect on Llama and Gemma, but additional seeds would tighten CIs.
- **Model coverage.** The modern grid spans Llama, Gemma, and two Qwen sizes. Some candidate models (Phi-4-mini, SmoLM3-3B) failed to load due to library/version mismatches and are reported separately rather than as nulls. A DeepSeek-R1-Distill run was abandoned because the reasoning trace format prevented clean decision parsing.
- **Defense evidence is strongest for Llama.** The local artifacts show Llama defenses working under both pools. They do not show Gemma 4 defense restoration, even though Gemma 4 itself is attack-susceptible.
- **Synthetic posts.** The generator-swap test substantially improves robustness against post-style critiques, but real social posts would further strengthen ecological validity.
- **Visible-history mediation.** The attack works through ordinary context accumulation. That is operationally important, but it is not evidence of a hidden internal-only mechanism.
- **Prompt sensitivity.** Earlier experiments showed that small changes to the final decision format can suppress or expose feed effects. Claims must be tied to the tested decision interface.

7 Conclusion

We have shown that adversarial ranked-feed exposure can significantly shift downstream decisions in susceptible modern LLM agents. The effect replicates across post generators, follows a dose-response curve, is asymmetric with respect to model defaults, and can be mitigated by simple feed-level defenses in the cleanest susceptible model. Other models exhibit saturated defaults, showing

that susceptibility is model-specific rather than universal. The activation-level signal that originally motivated the project is largely visible-history mediated and serves as a methodological warning: in multi-turn LLM-agent settings, naive random-CV probing overstates the “hidden mechanism” content.

The central contribution is that recommender systems act as a practical control surface for LLM agents, and that this steering is bounded by the model’s default: a one-sided feed tips a movable decision but does not overwrite a firmly held one.

In an age of agentic AI, every recommender silently authors every reply. The question is no longer whether models behave well; the question is who controls what they read just before they answer.

Reproducibility

All code, post pools, and per-rollout decision logs are released alongside the paper. The complete source, the agent and pool-generation code, and the analysis and figure scripts are available at <https://github.com/ranausmanai/recommenders-as-control-surfaces>; the five figures regenerate from the released decision-rollout files via these scripts (see Appendix A for the file map). The agent protocol uses standard HuggingFace Transformers and Ollama, with no gated weights and no non-public APIs, and random seeds are recorded with every rollout. The post pools are released under CC-BY 4.0 as the Hugging Face dataset `ranausmans/feed-injection-pool`, and the per-rollout decision logs as `ranausmans/feed-injection-rollouts`.

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A Software identifiers, file layout, and code locations

For reproducibility, this appendix lists the mapping between the human-readable condition names used throughout the paper and the software identifiers used in the released code and rollout records.

Condition identifiers. The following mapping is used in the `condition` field of every rollout record:

Paper label	Identifier in code and rollout records
random baseline	<code>organic_random</code>
recency baseline	<code>organic_recency</code>
light injection (1/5 adv.)	<code>light</code>
heavy injection (5/5 adv.)	<code>heavy</code>
balanced defense	<code>balanced</code>
disclosed heavy injection	<code>disclosed_heavy</code>
dose-response, $k/5$ adv.	<code>dose0, dose1, ..., dose5</code>

Post-pool file layout. The five remote-work post pools are released as JSON-Lines files under the Hugging Face dataset repository (the additional generalization-task pools follow the same schema):

File	Contents
<code>pool.jsonl</code>	500 Claude-generated organic posts (5 topics)
<code>adversarial_rto.jsonl</code>	50 Claude pro-return-to-office adversarial posts
<code>adversarial_pro_remote.jsonl</code>	50 Claude pro-remote adversarial posts (anti-direction control)
<code>pool_gemma.jsonl</code>	100 Gemma-generated organic posts (remote-work topic)
<code>adversarial_rto_gemma.jsonl</code>	50 Gemma pro-return-to-office adversarial posts

Rollout-record file layout. The 2,785 decision rollouts are released under the Hugging Face dataset `ranausmans/feed-injection-rollouts`. The headline cross-model attack data resides in `decision_shift_adv_modern.jsonl`; the generator-swap, anti-direction, and dose-response data resides in `decision_shift_followup.jsonl`; and the cross-task generalization grid (Section 4.5) resides in `decision_shift_tasks.jsonl`. The generalization tasks also use additional organic and adversarial post pools (`pool_<task>.jsonl`, `adversarial_<task>.jsonl`) released in the same dataset repository.

Analysis scripts. Figures 1–4 regenerate from the JSONL files via `notebooks/11_paper_figures.py`, and the cross-task generalization figure via `notebooks/13_task_figure.py`, in the companion GitHub repository.